

the system. Popular audio codecs include MP3 (MPEG Layer 3), AAC, and WMA (Windows Media Audio).

#### **4.4.5 Decibel**

A measure of sound intensity, calculated as the logarithmic value of a ratio of sound intensities. The basis is considered to be the threshold of hearing for a normal human ear. Every other sound is expressed in decibels in relation to this base sound intensity. 0 dB represents the threshold of hearing, while 120 is the higher level that the human ear can tolerate without feeling pain.

#### **4.4.6 Dynamic Range**

This refers to the range of loudness of sounds that can be detected by a recording device or reproduced by a speaker. This is measured as the difference in the highest and lowest sound, in decibels. The human ear has a dynamic range of 120 dB. A high dynamic range of recording or playback is representative of high-quality equipment.

#### **4.4.7 MIDI**

Musical Instrument Digital Interface is the default interface to connect any digital musical instrument to a sound card. This allows data to be transferred across the sound card and the device, and allows recording of any composition to the system. Each MIDI port can connect to up to 16 compatible devices simultaneously, enabling a single PC to control these devices. MIDI uses a standard set of codes that conveys information about how to produce the sound—rather than the sound itself—so MIDI files are small compared to the audio file for the same content. Since a MIDI file relies on the sound card's wavetable to create an output, the results may vary with the sound card. The 15 pin D-sub MIDI interface also doubles as a gameport to plug in compatible gamepads.

#### **4.4.8 Mono**

An audio stream that contains only a single data channel would be called Mono (for monaural).